

Control work 6, and the quarter review

Combinatorics, the binomial theorem and statistics in one paper, then the map of the whole quarter.

LESSON 77–78

2 × 40 MIN

ALGEBRA 11, NAZORAT ISHI 6

IGCSE E20 REVIEW

LESSON CLOCK

3

42'

10'

20'

10'

5'

Instructions	3 min
The paper	42 min
Answers	10 min
Rewrite	20 min
Concept map	10 min
Targets	5 min

LEARNING OBJECTIVES

- Apply the combinatorial and statistical methods under time.
- Decide whether order matters, unprompted.
- Build a concept map of the quarter.
- Set a personal target for Quarter IV.

TEXTBOOK

National	pp. 365–368
Cambridge	Core & Extended, pp. 425–428
Workbook	Control paper F

01

Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	From 9 people: a team of 4; then a captain, a vice-captain and two others	5	L60–62
2	How many arrangements has SAMARQAND? And with the two A's together?	5	L60–62
3	Find the term in x^4 in $(2x - 1)^7$	4	L63–64
4	Estimate the mean and standard deviation of a grouped table	6	L68–70
5	Draw a cumulative frequency curve and read off the median and quartiles	6	L65–67
6	Describe a given correlation and say why a prediction at $x = 40$ is invalid	4	L71–73

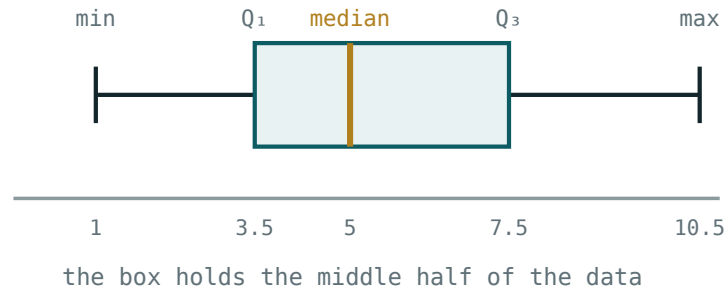
WHERE THE MARKS HIDE

Q1 gives one mark for saying whether order matters in each part. Q6 gives two words — type and strength — and two for the extrapolation reason. Five of thirty are for words, not numbers.

The concept map

Six boxes, links as sentences:

- **antiderivative** → **definite integral** — “Newton–Leibniz: $F(b) - F(a)$ ”
- **definite integral** → **area and volume** — “upper minus lower; $\pi \int y^2 dx$ ”
- **no antiderivative** → **trapezium rule** — “straight chords, error $\propto h^2$ ”
- **counting rules** → **$C(n,k)$** — “order matters, or it does not”
- **$C(n,k)$** → **binomial theorem** — “Pascal’s triangle is the coefficients”
- **data** → **centre and spread** — “mean and σ , or median and IQR”



The picture behind the last box.

Looking forward

Quarter IV is probability and distributions, then the Cambridge revision blocks on complex numbers and differential equations. The combinatorics of this quarter is what makes probability computable, and the integration is what solves the differential equations.

ONE HABIT TO CARRY FORWARD

Ask “does order matter?” before every counting question, and “is this the mean or the median’s job?” before every statistic. Both are one-sentence decisions that determine the whole answer.

02 Worked examples

EXAMPLE 1 Model answer, Q1: from 9 people, a team of 4; then a captain, vice-captain and two others.

① A team: order does not matter.
 $C(9,4) = 126$

② With roles: choose the captain, then the vice, then two others.
 $9 \times 8 \times C(7,2)$

③ $= 9 \times 8 \times 21 = 1512$

ANSWER

126; 1512

EXAMPLE 2 Model answer, Q2: arrangements of SAMARQAND.

- ① 9 letters with A×3.

$$\frac{9!}{3!} = 60\,480$$

- ② With the three A's together: treat them as one block.

$$7! = 5040$$

- ③ (With only two A's together the count differs.)

ANSWER

60 480; 5040 with all three A's together

EXAMPLE 3 Model answer, Q3: the term in x^4 in $(2x - 1)^7$.

- ① $7 - k = 4 \Rightarrow k = 3$

- ② $C(7,3) = 35$

- ③ $35 (2x)^4 (-1)^3$

- ④ $= 35 \times 16 \times (-1) x^4 = -560x^4$

ANSWER

$-560x^4$

03 Interactive model

Work Q1 and Q3 on the board, stating the order question aloud each time.

The quarter in ten questions

$\int_0^3 x^2 dx$:

9

27

3

6

Area between $y = 4x$ and $y = x^2$:

$\frac{16}{3}$

$\frac{32}{3}$

16

8

Volume from $y = \sqrt{x}$ on $[0, 4]$:

4π

8π

16π

2π

4 strips need how many ordinates?

3

4

5

8

A team of 4 from 9:

126

3024

36

24

Arrangements of SAMARQAND:

9!

$\frac{9!}{3!}$

$\frac{9!}{2!}$

7!

Term in x^4 of $(2x - 1)^7$:

$$560x^4$$

$$-560x^4$$

$$35x^4$$

$$-35x^4$$

Which average resists outliers?

the mean

the median

the mode

none

σ^2 equals:

$$\frac{\sum x}{n}$$

$$\frac{\sum x^2}{n} - \bar{x}^2$$

$$\sqrt{\sum x^2}$$

$$\frac{\sum (x - \bar{x})}{n}$$

Predicting outside the data range is:

interpolation

extrapolation

correlation

valid

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Concept map	Tushunchalar xaritasi	Карта понятий
Permutation	O'rin almashtirish	Перестановка
Combination	Kombinatsiya	Сочетание
Standard deviation	Standart chetlanish	Стандартное отклонение
Correlation	Korrelyatsiya	Корреляция
Self-assessment	O'z-o'zini baholash	Самооценка
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first question in any counting problem is:

how many?

does order matter?

is it a factorial?

is it large?

A grouped mean is:

exact

an estimate

wrong

the median

A definite integral's answer is:

a family

a number

a function

a constant of integration

Quarter IV begins with:

- integration
- probability
- vectors
- logarithms

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $C(9,4)$

ANSWER 126

2. $A(9,2)$

ANSWER 72

3. Arrangements of SAMARQAND

ANSWER 60 480

4. Term in x^4 of $(2x - 1)^7$

ANSWER $-560x^4$

5. $\int_0^3 x^2 dx$

ANSWER 9

6. Mean of 4, 6, 8, 10

ANSWER 7

7. Median of 4, 6, 8, 10, 30

ANSWER 8

Medium · the standard question

7 PROBLEMS

1. A captain, a vice and two others from 9

ANSWER 1512

2. Arrangements of SAMARQAND with the three A's together

ANSWER 5040

3. Area between $y = 4x$ and $y = x^2$

ANSWER $\frac{32}{3}$

4. Volume from $y = \sqrt{x}$ on $[0, 4]$

ANSWER 8π

5. σ of 2, 4, 4, 4, 5, 5, 7, 9

ANSWER 2

6. Grouped mean, midpoints 5,15,25,35, $f = 4,9,12,5$

ANSWER 21

7. Estimate $\int_0^4 x^2 dx$ with 4 strips

ANSWER 22

Hard · stretch and reason

7 PROBLEMS

1. From 7 men and 5 women, committees of 5 with at least 3 women

ANSWER 246

2. Term independent of x in $(2x - \frac{1}{x^2})^9$

ANSWER -672

3. Grouped σ with midpoints 5,15,25,35, $f = 4,9,12,5$

ANSWER ≈ 8.6

4. Area enclosed by $y = x^3$ and $y = x$

ANSWER $\frac{1}{2}$

5. $\int \frac{5x-1}{x^2-1} dx$

ANSWER $2\ln|x-1| + 3\ln|x+1| + C$

6. $\int x \cos x dx$

ANSWER $x \sin x + \cos x + C$

7. Why is a prediction at $x = 40$ invalid for data on $[1, 8]$?

ANSWER Extrapolation far outside the range

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter IV.

1. Rewrite in full every control-work question that lost a mark.
2. Finish the concept map with all six links written as sentences.
3. From 8 men and 6 women, how many committees of 5 contain at least 2 women? And find the term in x^3 of $(3x - 2)^6$.

4. Write your target for Quarter IV in one checkable sentence, and date it.